DYNAMIC CONTEXT-AWARE TRIES FOR KNOWLEDGE GRAPH-CONSTRAINED LARGE LANGUAGE MODEL GENERATION

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Outline

Introduction

- LLMs hallucinate, producing factually incorrect outputs
- KGQA requires answers that follow real graph structure
- Constrained decoding (GCR, DoG) guarantees validity but not relevance
- A 3-hop question on Freebase can produce ∼24,000 structurally valid paths, yet only one is correct

Statement of Problem

- GCR's trie is fixed before decoding starts, so it cannot adapt to question intent or the evolving chain
- The LLM must filter semantically irrelevant paths using internal knowledge, the same mechanism that causes hallucinations

Project Objectives (1/2)

- Characterize the permissiveness problem by measuring SIR of GCR's static KG-Tries on WebQSP
- Design a symbolic constraint oracle (TypeOracle) with range and type gates, avoiding embedding cost and threshold sensitivity

Project Objectives (2/2)

- Implement DCA-Trie v1 (static filtering at construction time, FNR $<$ 5%) and v2 (dynamic expansion per committed triplet)
- Evaluate both against GCR baseline on WebQSP: Hits@1, F1, structural faithfulness, SIR, trie size

Tools and Facilities

Software

- Python, PyTorch, HuggingFace
- GCR Framework
- Sentence-transformers (MiniLM)
- NetworkX, marisa_trie
- Freebase (WebQSP, CWQ)

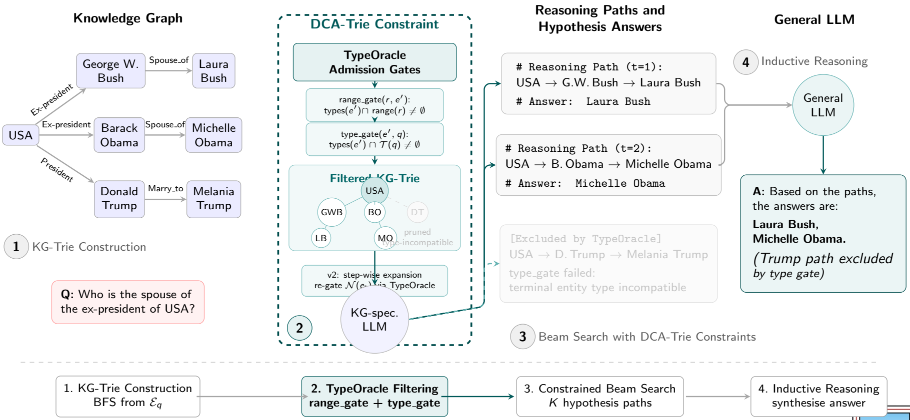
Hardware & Sources

- NVIDIA RTX 4090 (24 GB)
- ACM, arXiv, ACL Anthology

Methods

- Literature review and theoretical baseline
- Algorithm design and TypeOracle implementation
- Experimental benchmarking on WebQSP and CWQ

Methodology



Upgraded Constraint

Baseline (GCR/DoG):

$$W_{\text{val}}^{\text{GCR}}(t) = f(G, E_q)$$

Proposed (DCA-Trie):

$$W_{\text{val}}^{\text{DCA}}(t) = f(G, E_q, q, y_{<t})$$

Design Conditions

- **F (Faithfulness):** 100% structural grounding in KG triplets
- **R (Relevance):** Lower SIR than baseline
- **P (Recall):** FNR on gold paths $< 5\%$

Constrained Decoding in LLMs

At each token step t, the LLM produces logits $\mathbf{h}_t \in \mathbb{R}^{|V|}$ over the vocabulary V. A constraint mask $\mathbf{m}_t \in \{0,1\}^{|V|}$ zeros out invalid tokens:

$$\tilde{P}(x_t) = \mathrm{softmax}\big(\mathbf{h}_t + (1 - \mathbf{m}_t) \cdot (-\infty)\big)$$

- In GCR/DoG, $\mathbf{m}_t$ comes from a static trie built before decoding: $\mathbf{m}_t = f(G, E_q)$
- DCA-Trie conditions the mask on the question and partial output: $\mathbf{m}_t = f(G, E_q, q, y_{<t})$
- Only tokens corresponding to valid KG triples receive non-zero probability

TypeOracle: Gate Design

Two symbolic gates over KG ontology metadata, applied per candidate triple (e_t, r, e'):

Gate 1: Property Range Gate (every hop)

$$\text{range\_gate}(r, e') = \mathbf{1}\big[\text{types}(e') \cap \text{range}(r) \neq \emptyset\big]$$

Checks that the tail entity's types fall within the relation's expected range.

Gate 2: Answer Type Gate (terminal hop only)

$$\text{type\_gate}(e', q, h, L) = \mathbf{1}\big[\text{types}(e') \cap \mathcal{T}(q) \neq \emptyset\big]$$

Checks that the terminal entity matches the inferred answer type $\mathcal{T}(q)$.

Combined: admissible = range_gate $\wedge$ type_gate, with conservative fallback (admit when schema info is missing).

Diagnostic Metric: SIR

SIR measures the fraction of structurally valid paths that are semantically incompatible with the question.

$$\mathrm{SIR}(q, t) = \frac{\sum_{p \in \mathcal{P}(q,t)} \mathrm{irrelevant}(p, q)}{|\mathcal{P}(q, t)|}$$

Corpus average:

$$\overline{\mathrm{SIR}} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{L_q} \sum_{t=1}^{L_q} \mathrm{SIR}(q, t)$$

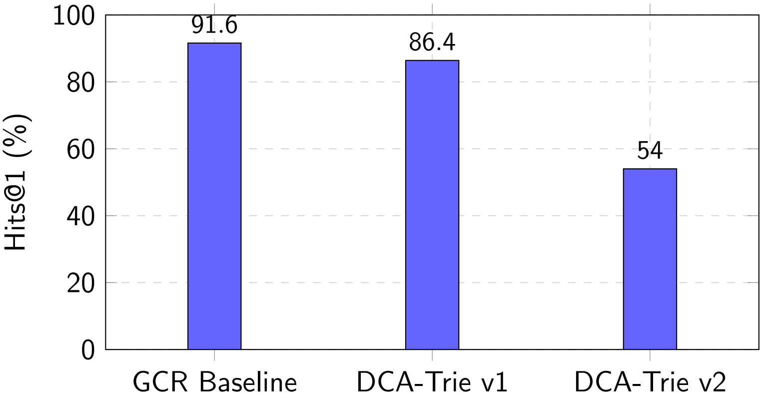
DCA-Trie v1: Static Filtering

- Filters BFS-enumerated paths through TypeOracle before building the prefix tree
- Preserves $O(d)$ trie lookup while pruning irrelevant branches offline

DCA-Trie v2: Dynamic Expansion

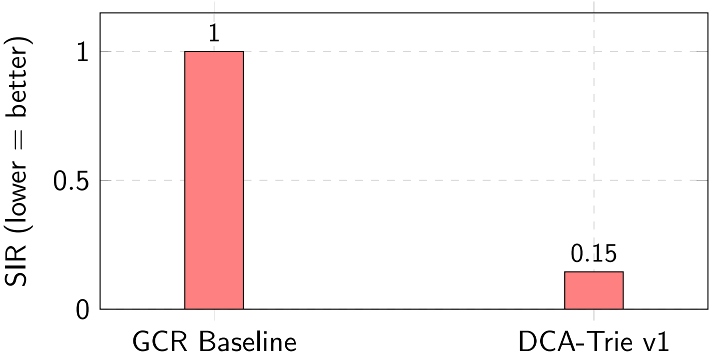
- Expands the trie step-by-step as entity tokens e_t are committed
- Applies range and type gate checks to $\mathcal{N}(e_t)$ during generation

Results: Hits@1 Comparison



WebQSP, 1,628 test queries, Llama-3.1-8B, beam $k=10$

Results: Constraint Quality



- 14.5% path reduction (4.10M $\rightarrow$ 3.51M paths)
- FNR: 3.3% (type gate), 2.9% (range gate)
- Latency overhead: $+0.5\%$

Key Contributions

Contribution	Before	After	Why It Matters
TypeOracle	Embedding + threshold τ	Two symbolic gates	No encoder, no τ, $O(1)$, deterministic
SIR Metric	Accuracy only	Semantic Irrelevance Ratio	Diagnoses constraint quality independently
Non-Monotone Finding	"Tighter $\rightarrow$ better" (GCR, DoG assumption)	Tightness $\neq$ accuracy (ΔAcc $= -5.2$pp)	First evidence constraint tightness is not a proxy for quality

Discussion

- **Non-monotone:** Tighter constraints do not guarantee better answers.
- **Four degradation mechanisms:** gold-path exclusion, beam competition, regex type-inference noise, precision–recall trade-off.
- **TypeOracle efficiency:** $O(1)$ per path, no encoder, no threshold, only $+0.5\%$ latency.
- **SIR fills a gap:** diagnoses constraint quality independently of answer accuracy.

Demonstration

Live demo: GCR baseline vs DCA-Trie v1
on a sample WebQSP question

Conclusion

- DCA-Trie v1 meets all three conditions: 100% faithfulness, 14.5% path reduction, FNR $< 5\%$.
- TypeOracle achieves meaningful tightening with $O(1)$ symbolic gates.
- Central finding: constraint tightness and accuracy are **non-monotone**, with Hits@1 dropping 5.2pp.
- SIR is introduced as the first diagnostic for constraint quality, decoupled from accuracy.

Recommendations

- Replace regex type classifier with LLM-based intent parser (85% $\rightarrow$ higher recall).
- Extend TypeOracle schema beyond 40 Freebase relations.
- Explore hybrid oracles (symbolic + embedding) to recover beam diversity.
- Evaluate on CWQ with $N \geq 500$ to reduce variance.

Thank You!!!

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